Lesson 4-3 · Session 1

Estimate with Benchmarks

Name _____ Date _____ Period _____

Maryland Standard 6.AT.4 · Learning target — I can estimate a percent by comparing an amount to benchmark percents.
Show a model, calculation, or explanation. Answer every part and label units when needed.

Exercises 1–12 earn 1 point for a correct response and 1 for supporting work. Spiral Review and Test Practice earn 1 point each.

Learn from a model

REMEMBER

- Benchmark percents: 10%, 25%, 50%, 75%, 100% are easy to estimate with mentally.
- Break a percent into benchmarks to estimate (e.g., 60% = 50% + 10%).
- Watch out — an estimate is not a guess. Name the benchmark you used.
- Check it — an estimate should be easy to defend in one sentence.

WORKED EXAMPLE

Estimating a Percent with Benchmarks A tablet holds 64 GB of storage and 47 GB are already used. About what percent of the tablet is full?

Step 1 Name the whole: all 64 GB stands for one hundred percent.

Step 2 Mark benchmarks: halfway is 32 GB, and three quarters of the way is 48 GB.

Step 3 Locate the used amount, 47 GB — it sits close to the three-quarter mark of 48 GB.

Step 4 Estimate with that benchmark, not with exact division.

Answer About three quarters — roughly seventy-five percent — of the tablet is full.

Practice

1. Estimate 48% of 80 using the benchmark 50%. _____

Answer _____

SHOW YOUR THINKING

2. Estimate 24% of 60 using the benchmark 25%. _____

Answer _____

SHOW YOUR THINKING

3. Explain how you could estimate 76% of a number using benchmarks.
Sentence Frame: "I know this because..."

Answer _____

EXPLAIN WITH MATHEMATICAL EVIDENCE

4. Name the benchmark percent (10%, 25%, 50%, 75%, or 100%) that each part is closest to.

a. 11 out of 20 is about _____ %
b. 19 out of 80 is about _____ %
c. 31 out of 40 is about _____ %
d. 9 out of 90 is about _____ %
e. 58 out of 60 is about _____ %

Answer _____

RECORD YOUR RESPONSE

5. Gas Tank. A gas tank holds 16 gallons and has 12 gallons in it. About what percent of the tank is full?

Answer _____

SHOW YOUR THINKING

6. Craft Beads. A jar holds 300 beads and 74 of them have been used. About what percent of the beads have been used?

Answer _____

SHOW YOUR THINKING

7. Homework Minutes. Devin planned 90 minutes of homework and has worked for 44 minutes so far. About what percent of the plan is done, and about what percent is left?

Answer _____

SHOW YOUR THINKING

8. Simone estimated that 7 out of 24 is about 50% because 7 is close to 10 and 24 is close to 20. Explain her mistake, then give a better estimate using a benchmark.

Answer _____

EXPLAIN WITH MATHEMATICAL EVIDENCE

9. A bag holds 60 counters and 29 are red. About what percent are red?

Answer _____ about what percent

SHOW YOUR THINKING

10. A drive is 200 miles and 52 miles are done. About what percent is done?

Answer _____ about what percent

SHOW YOUR THINKING

11. Name the Benchmark. A phone battery shows 78 of 80 units of charge. Name the closest benchmark percent and explain.

Answer _____ benchmark and why

EXPLAIN WITH MATHEMATICAL EVIDENCE

12. Find the Mistake. Asked to estimate 47 out of 64, Dana answered 47%. Explain the mistake.

Answer _____ the mistake

EXPLAIN WITH MATHEMATICAL EVIDENCE

Spiral review

13. From Lesson 4-2 Session 1. Write 0.25 as a fraction in lowest terms.

Answer _____

SHOW YOUR THINKING

14. From Lesson 4-1 Session 2. 150% of an amount is how many times the amount?

Answer _____

SHOW YOUR THINKING

15. From Lesson 3-6 Session 2. Convert 5500 millilitres to liters.

Answer _____

SHOW YOUR THINKING

Test practice

16. 50% of an amount is —

□ A. one fourth	□ B. one half
□ C. one tenth	□ D. all of it

17. 9 out of 40 is about —

□ A. 75%	□ B. 50%
□ C. 25%	□ D. 90%

18. 75% is the same as —

- □ A. seven tenths
- □ B. two thirds
- □ C. one third
- □ D. three fourths

Check and revise

CHECK YOUR MATHEMATICS	HOW DID IT GO?
Decide whether the question asks for an estimate or an exact result. Every part answered / Work shown / Units labeled when needed	I can do this: Independently / With a hint / With help Question to revisit: _____

One correction or question I want to discuss: _____

Identify the whole. Check your result against 0%, 50%, and 100% of that amount.